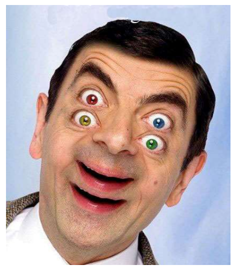
Introduction to Biophysics II: Quantum physics & biology

Lecture 20:
Vision: Quantum biology of retinal

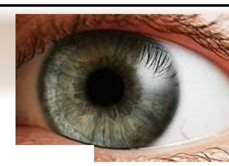


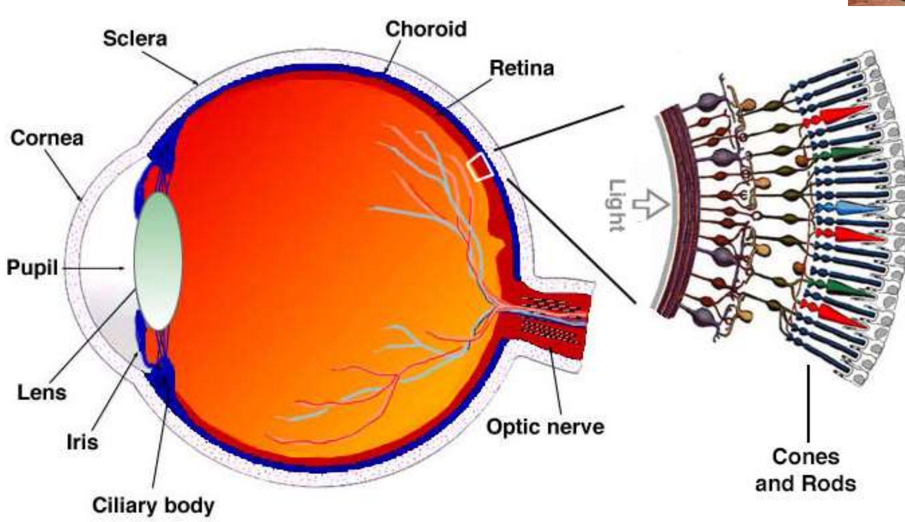
Reading:
Quantum effects in biology,
M. Moshedi, Y. Omar, GS Engel, M, Plenio
Chapter 11

Handbook of Molecular Biophysics. Methods and Applications G Groenhof, LV Schafer, M Boggio-Pasqua, MA Robb - 2009 - Weinheim: Wiley-VCH

1

An eye





https://www.blueconemonochromacy.org/how-the-eye-functions/

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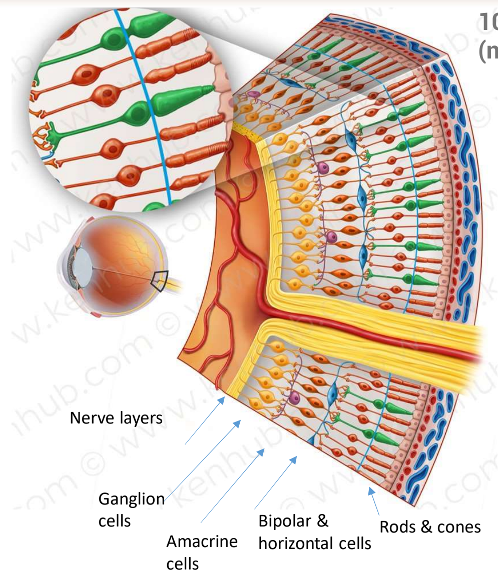
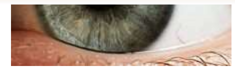
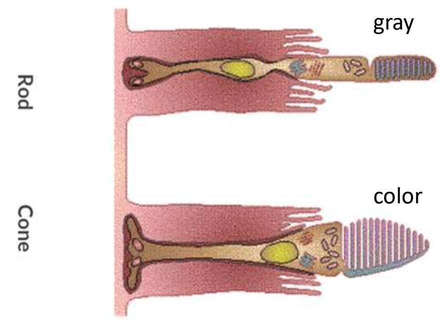
An eye



https://www.kenhub.com/en/library/anatomy/photoreceptors

100 megapixel camera (number varies depending on the source…)





Nerve layers:
initial image processing: shapes, bright spots, movement

By Henry Vandyke Carter - Henry Gray (1918) Anatomy of the Human Body, p. 1,016 (See "Book" section below)Bartleby.com: Gray's Anatomy, Plate 881, Public Domain, https://commons.wikimedia.org/w/index.php?curid=566820

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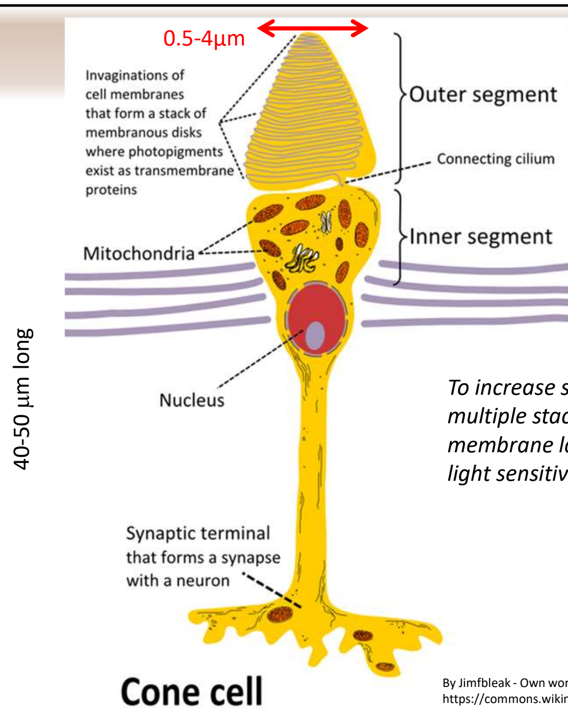
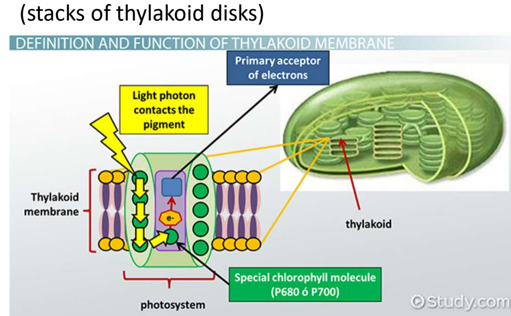
Vision vs photosynthesis



To increase sensitivity: multiple stacked membrane layers with light sensitive molecules

By Jimfbleak - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=4454798

Compare with **photosynthetic** membrane:
layers of thylacoid membrane with photosystems (stacks of thylakoid disks)



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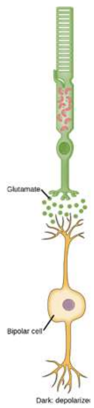
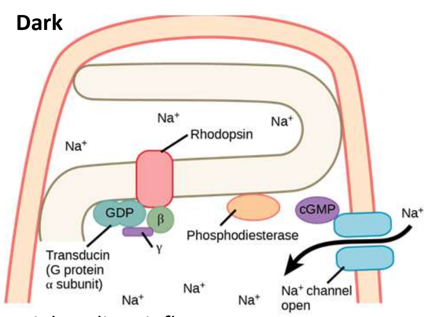
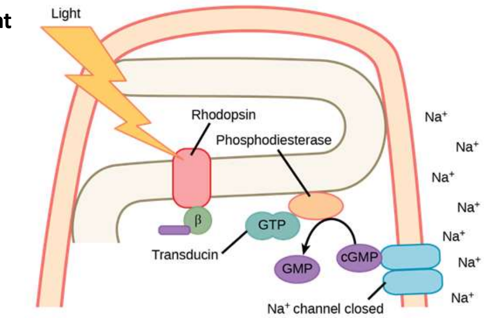
Rhodopsins

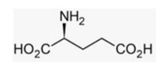




High sodium influx,
high neurotransmitter release



Low sodium influx, cell more negative,
reduce neurotransmitter release



Glutamate - amino acid
(excitatory neurotransmitter)

When light strikes rhodopsin, the G-protein transducin is activated, which in turn activates phosphodiesterase. Phosphodiesterase converts cGMP to GMP, thereby closing sodium channels. As a result, the membrane becomes hyperpolarized. The hyperpolarized membrane does not release glutamate (an excitatory neurotransmitter) to the bipolar cell.

https://courses.lumenlearning.com/wm-biology2/chapter/transduction-of-light/

An excitatory neurotransmitter excites or stimulates a nerve cell

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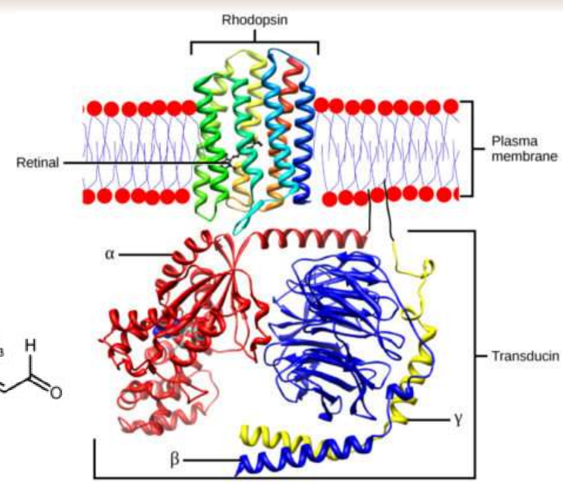
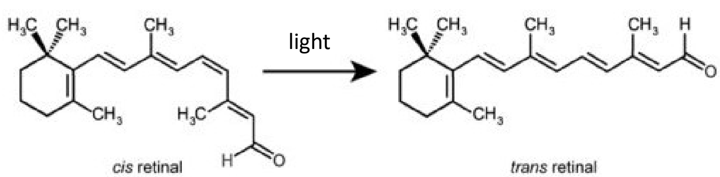
Rhodopsins

Rhodopsin (Rh)
- a membrane protein of the rod cells in the retina.
- Contains *retinal molecule*, as a chromophore that absorbs light and initiates signaling and *opsin* (transmembrane protein)
- Retinal has two conformations, cis and trans, switched by light





https://courses.lumenlearning.com/wm-biology2/chapter/transduction-of-light/

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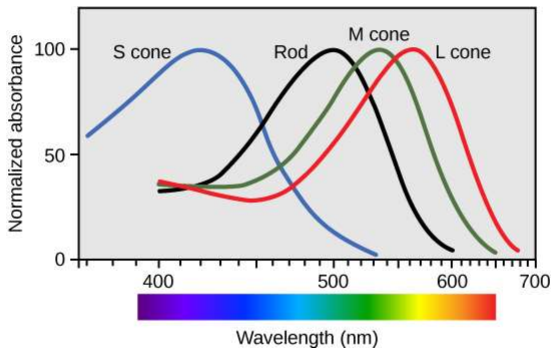
Spectral sensitivity



Photopsins (also known as Cone opsins): photoreceptor proteins in the cone cells of the retina that are the basis of color vision.
Rhodopsins: are used in night vision.

They are a combination of a chromophore and a protein

Note: there are several thousand different types of opsins (including non-visual opsins)

Most bind 11-cis *retinal*, optical differences stem from different binding in the protein pocket

Cone type	Name	Range	Peak wavelength[1][2]
S (OPN1SW) - "tritan", "cyanolabe"	β	400–500 nm	420–440 nm
M (OPN1MW) - "deutan", "chlorolabe"	γ	450–630 nm	534–545 nm
L (OPN1LW) - "protan", "erythrolabe"	ρ	500–700 nm	564–580 nm

In the human eye:
91 million rods (night vision)
4.5 million cones (color, day vision)
→ 100 Megapixel camera
(Highest density in the middle)

https://courses.lumenlearning.com/wm-biology2/chapter/transduction-of-light/

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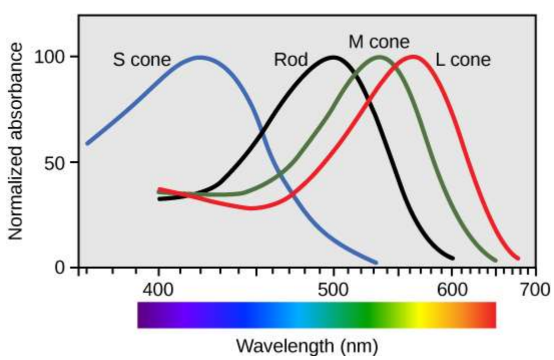
Spectral sensitivity

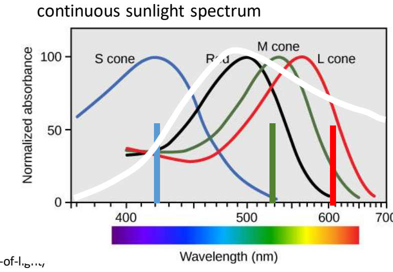


- We detect the ratio of RGB signals and deduce color.

- For monochromatic light, one can detect a difference in a few nanometers in wavelength.

- Color photography/displays excite different ratios of RGB channels with just three colors, the real spectrum of the original may be very different:

Just three spectral lines with appropriate
The intensity ratio can be perceived as 'white.'
continuous sunlight spectrum



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https://courses.lumenlearning.com/wm-biology2/chapter/transduction-of-l...

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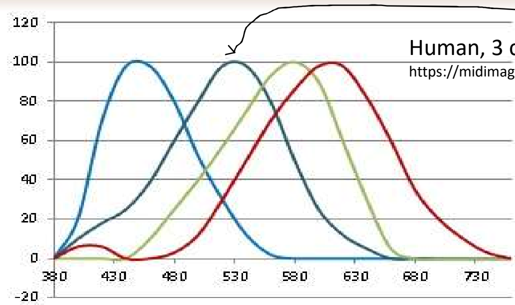
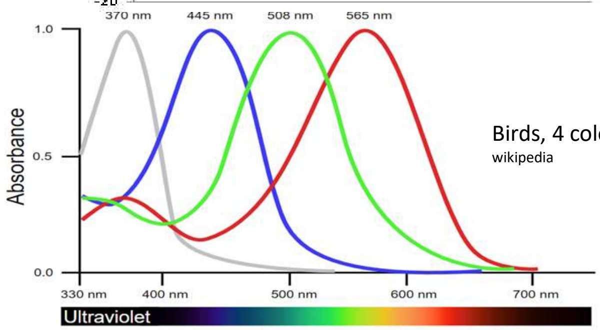
Bird vision: tetrachromacy



Human, 3 colors + black-white
https://midimagic.sgc-hosting.com/huvision.htm



Birds, 4 colors
wikipedia

Artist's model

https://www.boredpanda.com/human-vs-bird-vision/?utm_source=google&utm_medium=organic&utm_campaign=organic

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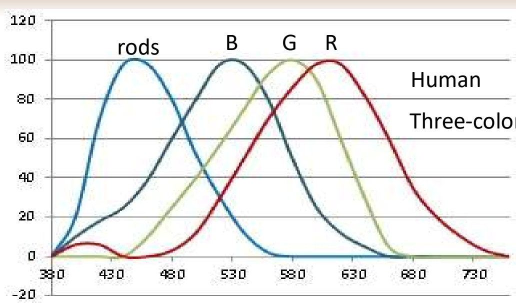
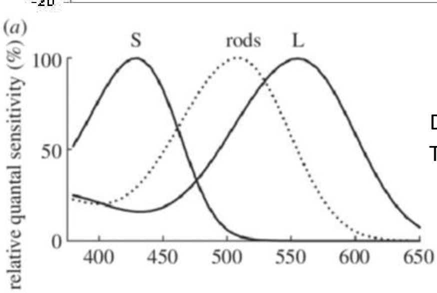
Dog's vision



Human
Three-color vision



Dog
Two-color vision

Bonsall et al
https://esajournals.onlinelibrary.wiley.com/doi/full/10.1890/13-0155.1

https://dog-vision.andraspeter.com/

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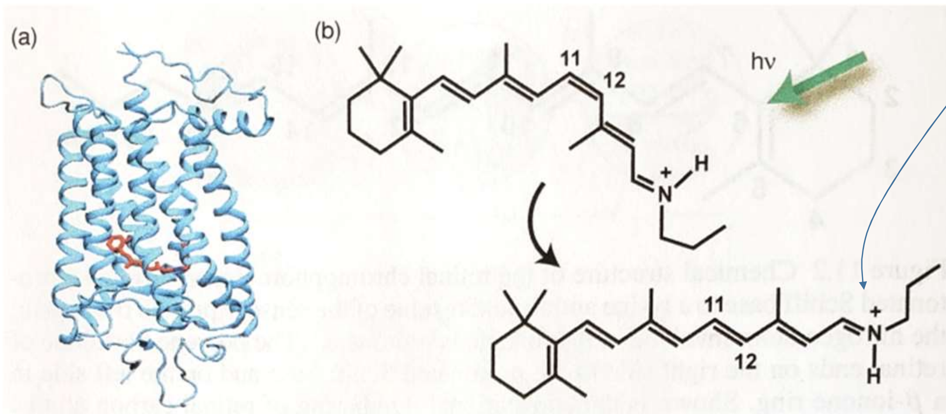
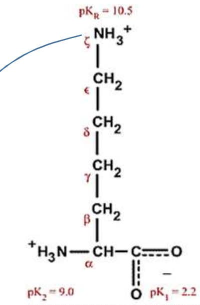
Rhodopsin (Rh) photodynamics





Lysine amino acid

Upon light excitation, the retinal molecule undergoes rapid (~50-500 fs) isomerization from 11-cis to all-trans
Note: retinal is shown bound to a lysine residue

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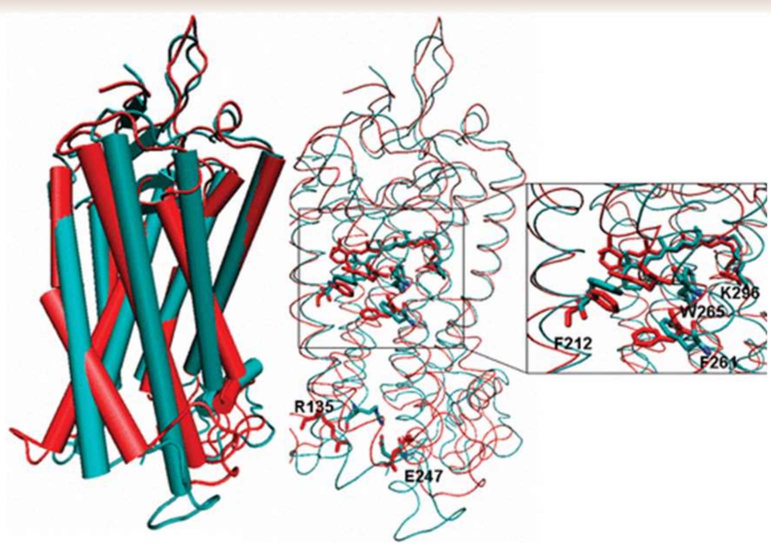
Rhodopsin (Rh) photodynamics



Structural changes in protein upon light excitation: LUMI (cyan) to META II (red) transition

J. Am. Chem. Soc. 2008, 130, 31, 10141–10149

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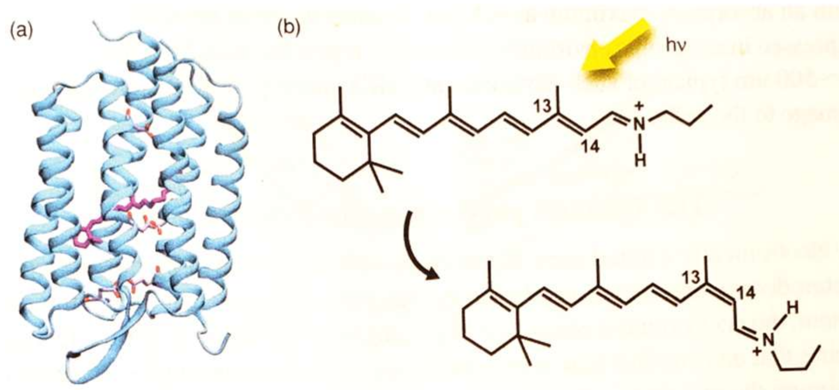
Bacteriorhodopsin (bR)

Rhodopsins are also found in bacteria: bacteriorhodopsin



In bacteria, they function in a somewhat reverse way; under light, they undergo all-trans →13-cis isomerization
This initiates proton pumping from the interior to the exterior of the bacterial cell wall to maintain cell potential, or to initiate phototaxis (to avoid light).

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Quantum physics of excited state dynamics

Use Born-Oppenheimer (BO) approximation, express Hamiltonian of the system as:

$$\hat{H}(\mathbf{r},\mathbf{R}) = \hat{T}_N(\mathbf{R}) + \hat{H}_{BO}(\mathbf{r},\mathbf{R})$$

r – electron coordinates
R - nuclear coordinates

Electronic Hamiltonian (depends of **R** parametrically)

Nuclear Kinetic energy

BO Hamiltonian: $$\hat{H}_{BO}(\mathbf{r},\mathbf{R}) = \sum_i^{N_{elec}}\left(-\frac{\hbar^2}{2m_e}\nabla_i^2\right) - \sum_i^{N_{elec}}\sum_A^{N_{nuc}}\frac{Z_A e^2}{4\pi\epsilon_0 r_{iA}} + \sum_i^{N_{elec}}\sum_{j>i}^{N_{elec}}\frac{e^2}{4\pi\epsilon_0 r_{ij}} + \sum_A^{N_{nuc}}\sum_{B>A}^{N_{nuc}}\frac{Z_A Z_B e^2}{4\pi\epsilon_0 R_{AB}}$$

H_{BO} = Electron Kinetic energy
\+ e-nuclear attraction
\+ e-e repulsion
\+ nuclear-nuclear (n-n) repulsion

Z – nuclear charge
R_{AB} – n-n distance
r_{ij} – e-e distance
r_{iA} – e-n distance

(we dropped kinetic energy of nuclei in H_{BO} – assume slow motion)

BO approximation → can separate nuclear and electronic wave functions: $\Phi_I\left(\vec{R}\right)\Psi_I\left(\vec{r},\vec{R}\right)$
where Φ(**R**) is the nuclear wavefunction, and Ψ_I is the electron wavefunction of the I^{th} electronic state at fixed nuclear coordinate ***R,*** given by the solution of electronic SE:

$$\hat{H}_{BO}(\mathbf{r},\mathbf{R})\Psi_I(\mathbf{r},\mathbf{R}) = E_I^{BO}(\mathbf{R})\Psi_I(\mathbf{r},\mathbf{R})$$

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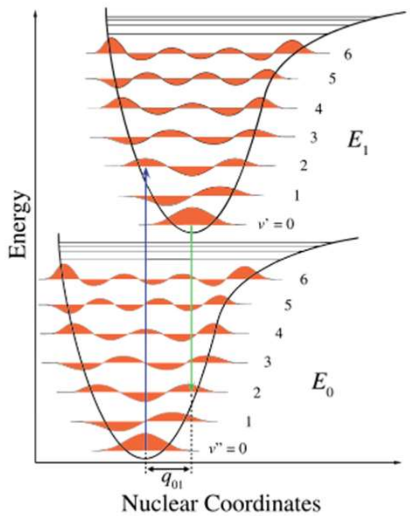
Quantum physics of excited state dynamics

From lecture 1- 5



The excited state (E_1) can decay to the ground state radiatively or nonradiatively

For rhodopsin to function, it needs to absorb light efficiently

→ Need a large transition dipole moment

→ Large transition dipole moment leads to short radiative lifetime

Recall (Lec 1-5):

Spontaneous emission rate $A = \dfrac{\omega_0^3 |\boldsymbol{\rho}|^2}{3\pi\epsilon_0 \hbar c^3}.$

$$|\rho|^2 \equiv |\vec{\mu}|^2 \equiv e\langle \psi_0 \,|\, \vec{r}_{10} \,|\, \psi_1 \rangle$$

In photosynthesis: needed to minimize nonradiative decay

In vision: need to minimize radiative decay

→ Minimize emission quantum yield $\Phi = \dfrac{k_{radiative}}{k_{radiative} + k_{nonradiative}}$

For efficient nonradiative decay need to maximize overlap between the excited and ground state *nuclear* wave functions:
Excited and ground state nuclear potential need to cross

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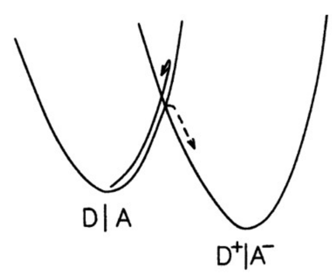
Electron transfer

Lecture 16-18

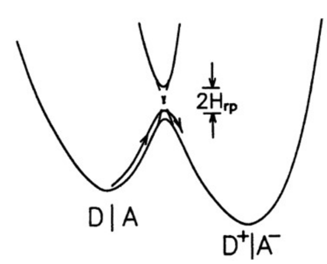
Recall: crossing nuclear surfaces leads to repulsion at cross-sectional area (similar to *exciton* theory):

Quantum-mechanical approach:

Weak coupling (*nonadiabatic, exchange of energy with bath)*
Previous derivation:
- perturbation theory only works with weak coupling.
- Electron states on donor and acceptor do not mix, electron is localized at all times



Strong coupling (*adiabatic, fast, no exchange with bath*)
- Perturbation theory does not work
- Approach is similar to *exciton* theory, will have new stationary states with electron shared between two molecules:



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Example: NaI light-induced dissociation

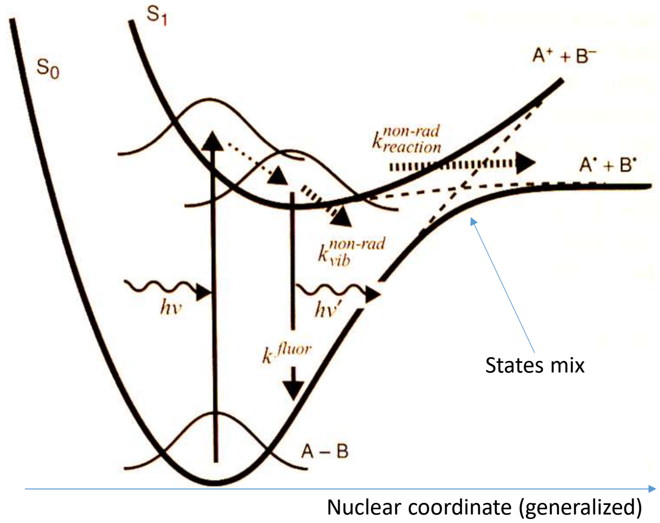
Example: NaI dissociation under light illumination

$$\mathrm{NaI} \xrightarrow{h\nu} \mathrm{Na}^+ + \mathrm{I}^-$$

Absorption of light: vertical transition (Franck-Condon principle, electron transition is fast, and nuclear coordinates do not change)

The excited state has 3 decay paths:
- fluorescence
- vibrational decay into the ground state
- crossing to the dissociated state

Thick solid line: adiabatic
dashed: diabatic (=nonadiabatic)



Note: Non-radiative transitions between electronic states with spin conservation are called *internal conversion.*
In that case, the transition is due to non-adiabatic coupling from the nuclear kinetic energy operator $T_N(\boldsymbol{R})$

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Quantum physics of excited state dynamics

Had:
Full Hamiltonian: $\hat{H}(\mathbf{r},\mathbf{R}) = \hat{T}_N(\mathbf{R}) + \hat{H}_{BO}(\mathbf{r},\mathbf{R})$

Basis functions

Full wave function: $\sum_I \Phi_I(\vec{R})\Psi_I(\vec{r},\vec{R})$, where Ψ_I is solution to $\hat{H}_{BO}(\mathbf{r},\mathbf{R})\Psi_I(\mathbf{r},\mathbf{R}) = E_I^{BO}(\mathbf{R})\Psi_I(\mathbf{r},\mathbf{R})$

Nuclear w.f. → expansion coefficients

$$\hat{H}\sum_I \Phi_I(\vec{R})\Psi_I(\vec{r},\vec{R}) = E\sum_I \Phi_I(\vec{R})\Psi_I(\vec{r},\vec{R})$$

$$\left(\hat{T}_N + \hat{H}_{BO}\right)\sum_I \Phi_I\Psi_I = E\sum_I \Phi_I\Psi_I$$

$$\hat{T}_N(\vec{R}) = \sum_A^{N_{nuc}}\left(-\frac{\hbar^2}{2M_A}\nabla_A^2\right)$$

Multiply by $\langle\Psi_J|\times$ and integrate over electronic coordinates *r*:

$$\left\langle \Psi_J \middle| \hat{T}_N \middle| \sum_I \Phi_I\Psi_I \right\rangle_r + \left\langle \Psi_J \middle| \hat{H}_{BO} \middle| \sum_I \Phi_I\Psi_I \right\rangle_r = E\left\langle \Psi_J \middle| \sum_I \Phi_I\Psi_I \right\rangle_r$$

$$\left\langle \Psi_J \middle| \hat{T}_N \middle| \sum_I \Phi_I\Psi_I \right\rangle_r + \left\langle \Psi_J \middle| \hat{H}_{BO} \middle| \sum_I \Phi_I\Psi_I \right\rangle_r = E\Phi_J$$

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Quantum physics of excited state dynamics

$$\left\langle \Psi_J \middle| \hat{T}_N \middle| \sum_I \Phi_I \Psi_I \right\rangle_r + \left\langle \Psi_J \middle| \hat{H}_{BO} \middle| \sum_I \Phi_I \Psi_I \right\rangle_r = E\Phi_J$$

$$\hat{T}_N(\vec{R}) = \sum_A^{N_{nuc}} \left(-\frac{\hbar^2}{2M_A} \nabla_A^2 \right)$$

$$\sum_I \Phi_I(\vec{R}) \Psi_I(\vec{r}, \vec{R})$$

$$= \sum_I \left\langle \Psi_J \middle| \hat{H}_{BO} \middle| \Phi_I \Psi_I \right\rangle_r = \sum_I E_I \Phi_I \left\langle \Psi_J \mid \Psi_I \right\rangle_r = E_J \Phi_J$$

Electronic energy

$$= \left\langle \Psi_J \middle| \sum_A \frac{-\hbar^2 \nabla_A \nabla_A}{2M_A} \middle| \sum_I \Phi_I \Psi_I \right\rangle_r = -\hbar^2 \sum_A \frac{1}{2M_A} \sum_I \left\langle \Psi_J \middle| \nabla_A \nabla_A \middle| \Phi_I \Psi_I \right\rangle_r$$

$$= -\hbar^2 \sum_A \frac{1}{2M_A} \sum_I \left\langle \Psi_J \middle| \nabla_A \middle| \Psi_I \nabla_A \Phi_I + \Phi_I \nabla_A \Psi_I \right\rangle_r$$

$\hat{T}_N$ Ignore second derivatives over R for electronic

$$= -\hbar^2 \sum_A \frac{1}{2M_A} \sum_I \left\langle \Psi_J \mid \Psi_I \nabla_A^2 \Phi_I + 2\nabla_A \Phi_I \nabla_A \Psi_I + \Phi_I \nabla_A^2 \Psi_I \right\rangle_r$$

$$\approx \hat{T}_N \Phi_J - \hbar^2 \sum_A \frac{1}{M_A} \sum_I \left\langle \Psi_J \mid \nabla_A \mid \Psi_I \right\rangle_r \nabla_A \Phi_I$$

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Quantum physics of excited state dynamics

$$\left\langle \Psi_J \middle| \hat{T}_N \middle| \sum_I \Phi_I \Psi_I \right\rangle_r + \left\langle \Psi_J \middle| \hat{H}_{BO} \middle| \sum_I \Phi_I \Psi_I \right\rangle_r = E\Phi_J$$

second derivatives of electronic wave functions with respect to **R** are neglected (but can also keep them)

$$\boxed{\hat{T}_N \Phi_J} - \boxed{\hbar^2 \sum_A \frac{1}{M_A} \sum_I \left\langle \Psi_J \mid \nabla_A \mid \Psi_I \right\rangle_r \nabla_A \Phi_I} + \boxed{E_J \Phi_J} \approx E\Phi_J$$

Kinetic energy of nuclei

The nonadiabatic coupling added to electronic states by nuclear motion Φ(**R**)

Electronic energy

Velocity of nucleus

(Recall Momentum: $\hat{p} = \frac{\hbar}{i}\nabla$)

$$\left\langle \Psi_J(\mathbf{r},\mathbf{R}) \middle| \hat{T}_N(\mathbf{R}) \middle| \Psi_I(\mathbf{r},\mathbf{R}) \Phi_I(\mathbf{R}) \right\rangle_r \simeq -i\hbar \sum_A d_{JI}^A(\mathbf{R}) \boxed{\frac{\hbar}{iM_A} \nabla \Phi_I(\mathbf{R})}$$

Integration over electronic coordinates

"Non-adiabatic coupling vectors" : $d_{JI}^A(\mathbf{R}) = \langle \Psi_J(\mathbf{r},\mathbf{R}) | \nabla_A | \Psi_I(\mathbf{r},\mathbf{R}) \rangle_r$

Handbook of Molecular Biophysics. Methods and Applications G Groenhof, LV Schafer, M Boggio-Pasqua, MA Robb - 2009 - Weinheim: Wiley-VCH

20

Regulation of photochemical process

Photoinduced processes in proteins are regulated according to function
Functions:
- Light-harvesting (photosynthetic antenna)
- Luminescence (like in green fluorescence protein)
- Photoactivation (as in rhodopsins)

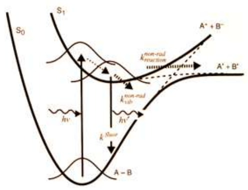
Common motif of Schiff base

Common requirement: strong photoabsorption

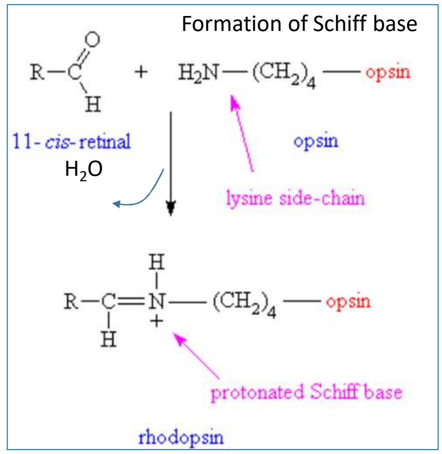
For rhodopsin, a strong transition dipole moment stems from the *Schiff base* formed by lysine bound to retinal

Specific for photoactivation: rapid non-radiative decay, need to cross to the ground state efficiently.

However, disorganized motion on the excited state surface would not necessarily lead to a crossing point. Need to trigger *very fast, **specific*** motion in the excited state, leading to the crossing point!



Protein environment helps to funnel nuclear motion in a desired direction



https://www.ch.ic.ac.uk/vchemlib/mim/bristol/retinal/retinal_text.htm

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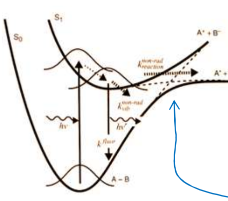
Potential energy crossing and conical intersection



Photodissociation of NaI depicts a prototype of the non-radiative process in chromophores.
This process is one-dimensional – **R** is just the distance between two atoms, Na and I.
Separation beyond the "avoided" energy crossing (AEC) leads to dissociation into neutral Na and I.

Note: BO energies of two states with the same spin and spatial symmetry wave functions *do not cross*.

The nonadiabatic term: $d^A_{JI}(\mathbf{R}) = \langle\Psi_J(\mathbf{r},\mathbf{R})|\nabla_A|\Psi_I(\mathbf{r},\mathbf{R})\rangle_r$
Contributes to the level of repulsion in THIS region (recall ET).

$$\langle\Psi_J(\mathbf{r},\mathbf{R})|\hat{T}_N(\mathbf{R})|\Psi_I(\mathbf{r},\mathbf{R})\rangle_r \simeq -i\hbar\sum_A^{N_{nuc}} d^A_{JI}(\mathbf{R})\,\frac{\hbar}{iM_A}\nabla_A$$

Non-adiabatic coupling vector can be rewritten as:

$$\langle\Psi_J(\mathbf{r},\mathbf{R})|\nabla_A|\Psi_I(\mathbf{r},\mathbf{R})\rangle_r = \frac{\langle\Psi_J(\mathbf{r},\mathbf{R})|[\nabla_A,\ \hat{H}_{BO}(\mathbf{r},\mathbf{R})]|\Psi_I(\mathbf{r},\mathbf{R})\rangle_r}{E_I^{BO}(\mathbf{R}) - E_J^{BO}(\mathbf{R})}$$

The denominator goes to infinity at the crossing coordinate **R,** leading to large splitting and rapid transfer from the bound state to the dissociated state.

This is a one-dimensional analog of the process in retinal: retinal is a more complex system with *many* nuclear coordinates, and for efficient transfer, all the coordinates must change in concert to enhance the transition to the product state.
Yet many organic molecules exhibit very small fluorescence quantum yield, suggesting effective crossing pathways.

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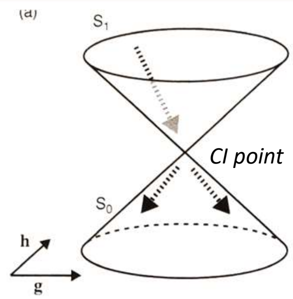
Potential energy crossing and conical intersection

Crossing in retinal occurs in ~50fs-200fs – comparable with the vibrational periods of molecules. This fast process can be explained through a non-adiabatic transition at *"conical intersection" (CI).*
There are ($3N_{nuc}$-6) degrees of freedom for complex molecular vibrations. For 2 dimensions (two-dimensional subspace), the nuclear potential energy surfaces of two states (ground and excited, S_0 and S_1) touch each other at one CI point.
The small energy gap between the BO surfaces near CI drastically accelerates the non-adiabatic transition.
For simplicity, consider the transition between excites and ground states defined as before: $\hat{H}_{BO}(\mathbf{r},\mathbf{R})\Psi_I(\mathbf{r},\mathbf{R}) = E_I^{BO}(\mathbf{R})\Psi_I(\mathbf{r},\mathbf{R})$



For I=0 and I=1. Can transform this via a unitary transformation to *diabatic* form, where the non-adiabatic coupling is minimized to about zero:

Diabatic form ˜: → $\tilde{\Psi}_{1,0}(\mathbf{r},\mathbf{R}) = \hat{U}(\mathbf{R})\Psi_{1,0}(\mathbf{r},\mathbf{R})$

$$\langle\tilde{\Psi}_1(\mathbf{r},\mathbf{R})|\hat{T}_N(\mathbf{R})|\tilde{\Psi}_0(\mathbf{r},\mathbf{R})\rangle_r \simeq 0$$

A unitary transformation is a transformation that preserves the inner product

Note: for 1D diatomic reactions, determining diabatic wave functions of ground and excited states is straightforward. It is not, however, possible to eliminate it to 0 for polyatomic molecules, since *all* coordinate components should vanish simultaneously in $d_{JI}^A(\mathbf{R}) = \langle\Psi_J(\mathbf{r},\mathbf{R})|\nabla_A|\Psi_I(\mathbf{r},\mathbf{R})\rangle_r$

Workaround: choose a reaction coordinate along which coupling is minimized

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Potential energy crossing and conical intersection

In the modified basis of diabatic electronic wavefunctions, Hamiltonian $H_{BO}(\mathbf{r},\mathbf{R})$ is no longer diagonal.
Express it as:

$$\hat{H}_{BO}(\mathbf{r},\mathbf{R}) = \begin{pmatrix} \tilde{W}_1(\mathbf{R}) & \tilde{V}(\mathbf{R}) \\ \tilde{V}(\mathbf{R}) & \tilde{W}_0(\mathbf{R}) \end{pmatrix}$$

Where diagonal elements can be understood as potential energies of new diabatic states.
Unlike adiabatic energies, diabatic energies can cross.
The off-diagonal elements are similar to interactions between individual chromophores in homodimeric excitons.

→ The energy curves of the adiabatic states with the same spatial and spin symmetries are "avoided" with a minimum energy gap $2V(\mathbf{R})$.
It is possible to find a point $\mathbf{R}=\mathbf{R}_{CI}$, where:

$$\tilde{W}_1(\mathbf{R}_{CI}) - \tilde{W}_0(\mathbf{R}_{CI}) = 0$$
$$\tilde{V}(\mathbf{R}_{CI}) = 0$$

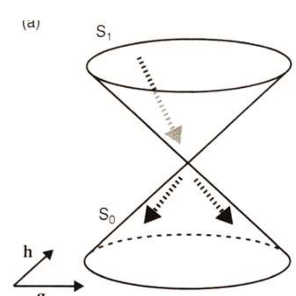
In that case, diabatic energies coincide, and there is no repulsion between diabatic states.
Inverse unitary transformation leaves these states degenerate (same energies), i.e.

$$E_1^{BO}(\mathbf{R}_{CI}) - E_0^{BO}(\mathbf{R}_{CI}) = 0$$

Taylor expansion around $\mathbf{R}_{CI}$ determines two directions **h,g** along which potential energies separate:

$$\mathbf{g} \equiv \nabla_{\mathbf{R}}[E_1^{BO}(\mathbf{R}) - E_0^{BO}(\mathbf{R})]|_{\mathbf{R}=\mathbf{R}_{CI}}$$
$$\mathbf{h} \equiv \nabla_{\mathbf{R}}\langle\Psi_1(\mathbf{r},\mathbf{R})|\hat{H}_{BO}(\mathbf{r},\mathbf{R})|\Psi_0(\mathbf{r},\mathbf{R})\rangle_r|_{\mathbf{R}=\mathbf{R}_{CI}}$$

h = *"gradient difference vectors"*
g = *"derivative coupling vector"*



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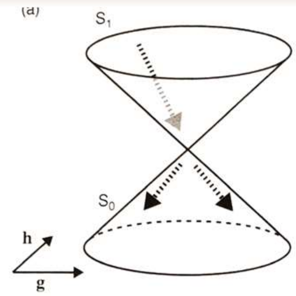
Potential energy crossing and conical intersection

Problem: the transition rate of non-radiative decay through conical intersection (CI region) is not properly described by perturbation theory (like Fermi's golden rule), because:
- large non-adiabatic coupling,
- Close proximity of two energy levels,
- Explicit dependence on nuclear motion on BO energy surfaces



For proper account:
- nuclear wave packet motion and the electronic transition need to be described by the use of the time-dependent Schrödinger equation for nuclear motion with multi-electronic state coupling for non-adiabatic interaction.

However: the task becomes too time-consuming for many-atom molecules

Workaround: semi-classical treatment

(i) Nuclear motion on a single potential surface can be described classically using Newton's equation:

$$M_A \ddot{\mathbf{R}}_A^I(t) = -\frac{\partial E_I^{BO}(\mathbf{R})}{\partial(\mathbf{R}_A)} \quad \leftarrow$$

ma=F, or ma=-grad(energy), for atom A

(ii) Electronic transition between the excited and ground electronic states is described with relevant coupling:

$$\langle \Psi_J(\mathbf{r},\mathbf{R})|\hat{T}_N(\mathbf{R})|\Psi_I(\mathbf{r},\mathbf{R})\rangle_r \simeq -i\hbar\,\mathbf{d}_{JI}(\mathbf{R})\cdot\dot{\mathbf{R}}(t)$$

(scalar product of coupling vector and velocity)

$$\hat{v} = \frac{\hat{p}}{M} = \frac{-i\hbar\nabla}{M}$$

Note: we replaced velocity operator with classical velocity: $\langle \Psi_J(\mathbf{r},\mathbf{R})|\hat{T}_N(\mathbf{R})|\Psi_I(\mathbf{r},\mathbf{R})\rangle_r \simeq -i\hbar \sum_A^{N_{nuc}} d_{JI}^A(\mathbf{R}) \boxed{\frac{\hbar}{iM_A}\nabla_A}$

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Potential energy crossing and conical intersection

Case of 1 dimension:

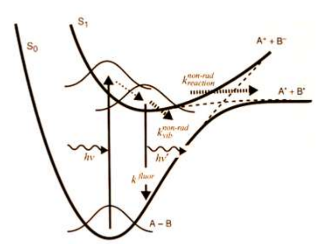
Landau-Zener (LZ) theory (1932) with assumptions:
- The velocity $\dot{\mathbf{R}}(t)$ is constant
- The energy difference between two diabatic states varies linearly with time: $\Delta\tilde{W}(R(t)) = \tilde{W}_1(R(t)) - \tilde{W}_0(R(t))$
- Off-diagonal element of two diabatic states is constant, V(**R**(t)) =V

Then it can be shown that the transition probability between two *adiabatic* states is:

$$P_{LZ}^{ad} = \exp\left[-\frac{\pi}{4\hbar}\left|\frac{\Delta E^{BO}(R_c)}{\dot{R}d_{10}(R_c)}\right|\right]$$

Where $\Delta E^{BO}(R_c) = E_1^{BO}(R_c) - E_0^{BO}(R_c) = 2\tilde{V}$ = energy difference between the two adiabatic states at crossing point R_c

Conclusion: Probability increases when
- Energy difference ΔE^{BO} becomes smaller
- Nuclear velocity is large
- Non-adiabatic coupling is large



Case of multidimensional potential
- Potential energy surface is complex
- V(R) depends strongly on nuclear coordinates

→ Assumptions of LZ theory are violated

General conclusions, however, are the same

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Potential energy crossing and conical intersection

Case of multiple dimensions:
Often treated using *Tully's fewest-switches algorithm (1990):*
- Probability of hopping between surfaces is determined by numerically solving the time-dependent Schrödinger equation

In the case of two electronic states, we construct the wave function as a superposition of two adiabatic ones (ground and excited states):

$$\Psi = c_0\psi_0\left(\vec{r},\vec{R}\right) + c_1\psi_1\left(\vec{r},\vec{R}\right)$$

Semi-classical couplings

And solve the time-dependent Schrödinger equation:

$$i\hbar\begin{pmatrix}\dot{c}_1(t)\\ \dot{c}_0(t)\end{pmatrix} = \begin{pmatrix}E^1_{BO}(\mathbf{R}(t)) & -i\hbar\mathbf{d}_{10}\cdot\dot{\mathbf{R}}(t)\\ -i\hbar\mathbf{d}_{01}\cdot\dot{\mathbf{R}}(t) & E^0_{BO}(\mathbf{R}(t))\end{pmatrix}\begin{pmatrix}c_1(t)\\ c_0(t)\end{pmatrix} \qquad \longleftarrow \qquad i\hbar\frac{\partial\Psi}{\partial t} = \mathrm{H}\Psi$$

(Similar to what we did for light absorption, Lectures 01-05)

The Schrödinger equation is solved along a classical trajectory **R**(*t*) on the potential energy surface of the representative state, whose population is supposed to be dominant.
If state 1 (excited) state is initially populated, the probability of "hopping" to the 0 (ground) state in time Δt is:

$$P^{ad}_{Tully} = \frac{-\dot{\rho}_{11}\Delta t}{\rho_{11}}$$

Where Δt is one time-step of numerical simulation and $\rho_{11} = |c_1|^2$ is the population of state 1

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Potential energy crossing and conical intersection

Tully's fewest-switches algorithm (1990):
Computer simulation loop:
0) start at time t=0, $c_1(0)=1$, $c_0(0)=0$, pick small enough time step Δt

LOOP over time *t*

i) Find derivatives of $\partial c_1/\partial t$ and $\partial c_1/\partial t$ at time *t* solving time-dependent Schrödinger equation:

$$i\hbar\begin{pmatrix}\dot{c}_1(t)\\ \dot{c}_0(t)\end{pmatrix} = \begin{pmatrix}E^1_{BO}(\mathbf{R}(t)) & -i\hbar\mathbf{d}_{10}\cdot\dot{\mathbf{R}}(t)\\ -i\hbar\mathbf{d}_{01}\cdot\dot{\mathbf{R}}(t) & E^0_{BO}(\mathbf{R}(t))\end{pmatrix}\begin{pmatrix}c_1(t)\\ c_0(t)\end{pmatrix}$$

ii) Find the probability of transition during this time step,

$$P^{ad}_{Tully} = \frac{-\dot{\rho}_{11}\Delta t}{\rho_{11}} \qquad \text{, where } \rho_{11} = |c_1|^2$$

iii) Compare that probability with a random number between 0 and 1,
If the random number is $<P_{tully}$ – there was a hop to state 0

iv) If not - Increase time by Δt: $t = t + \Delta t$, repeat starting from step i)

The transfer rate-determining process is reaching the CI point (bottleneck).
Once the system is at CI, transfer between states is inevitable and fast. (similar to chemical reactions, activated complex)

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Potential energy crossing and conical intersection

Example

Consider two steps - motion to CI and transition at CI:
Two steps are sequential, so:

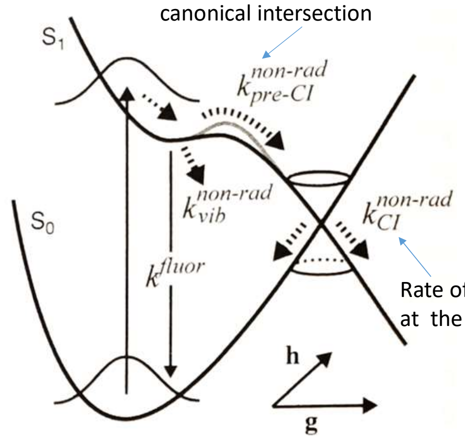
$$\frac{1}{k^{non-rad}_{reaction}} \approx \frac{1}{k^{non-rad}_{pre-CI}} + \frac{1}{k^{non-rad}_{CI}}$$

However: $k^{non-rad}_{CI} \gg k^{non-rad}_{pre-CI}$

$$k^{non-rad}_{reaction} \simeq k^{non-rad}_{pre-CI}$$

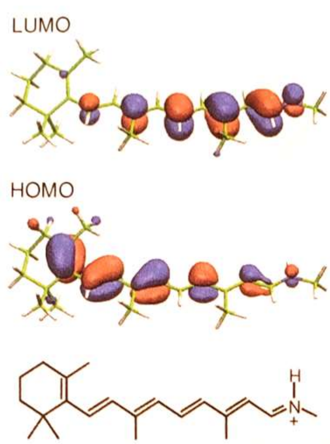
Process competing with fluorescence is the motion toward CI

The crucial process is the evolution of the excited state after the system is excited.
Vibrations initiated after Franck-Condon transitions must be optimized to bring nuclei to the configuration of CI in just a few vibrations.
A large number of vibrations must be synced to make that process faster. Protein structure defines vibrations!

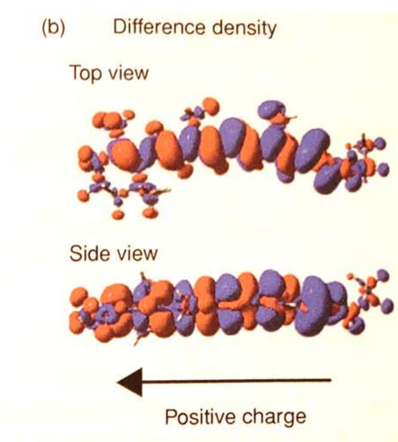


Electronic structure of retinal

Example

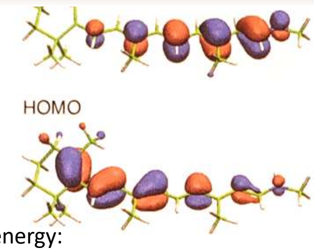


Characteristic lobes in wave function stem from π-orbitals



Very large transition dipole moment: ~10 Debye
About twice as large as that of Chl *a*
How much stronger is absorption?

Spectral tuning



- All retinal proteins enclose the same protonated-Schiff-base (PSB) retinal chromophore

- Spectral tuning is achieved by changes in protein environment that affect transition energy

(i) The electrostatic field of the surrounding protein affects the chromophore's transition energy: Stark effect (often called "electrochromic shift" by biologists, or solvation by chemists). The molecule possesses a strong permanent dipole moment, which is *different* in ground and excited states. This shifts the optical transition energy due to the different energy of the dipole in the electric field:

$$h\Delta\nu = \vec{\mu}_1^{perm}\cdot\vec{E} - \vec{\mu}_0^{perm}\cdot\vec{E} = \Delta\vec{\mu}_{01}^{perm}\cdot\vec{E}$$

(oversimplified, must take into account also screening effects and inhomogeneity of electric field)

(ii) Conformational changes of PBS retinal chromophore due to the protein binding pocket. The Polyene chain of the chromophore can be deformed torsionally in the ground state. Torsion around double bonds causes a decrease in transition energy; torsion around single bonds causes an increase.

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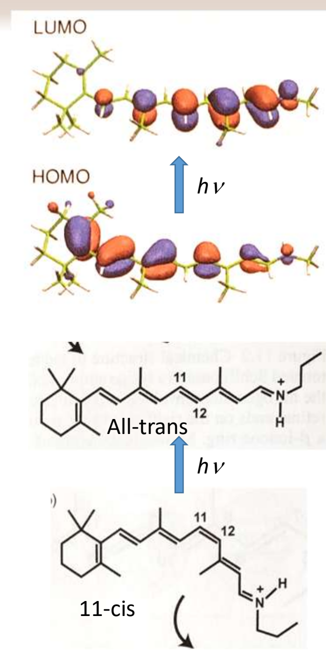
Photoisomerization



Excitation of retinal leads to a "hot" nonequilibrated S_1 state and induces strong torsional motion in a polyene chain.

- S_1 state possesses ionic valence-bond configuration and reduction of overlap between 2p orbitals constituting molecular π-orbitals at double bonds, and one of the double bonds undergoes torsion towards a 90^0 conformation.
- A remarkable charge separation, "*sudden polarization"* in the excited "twisted" conformation, brings the excited and ground state energy close together, forming a conical intersection (CI)
- Excited state decays to ground state, where perpendicular geometry relaxes to planar branching either to the original isomeric state of the double bond or to the alternate isomeric state.

Photoisomerization in rhodopsin is one of the fastest known molecular reactions in nature.
Excited state decays in 50 fs.
→ Fluorescence is almost completely quenched (quantum yield ~10^{-5}).
→ Photoactivation yield is 0.68
Note: pump-probe experiments reveal *coherent* wavepacket dynamics from FC region to photoproduct, even in the presence of surrounding protein noise → , motion occurs along a single path straight to CI

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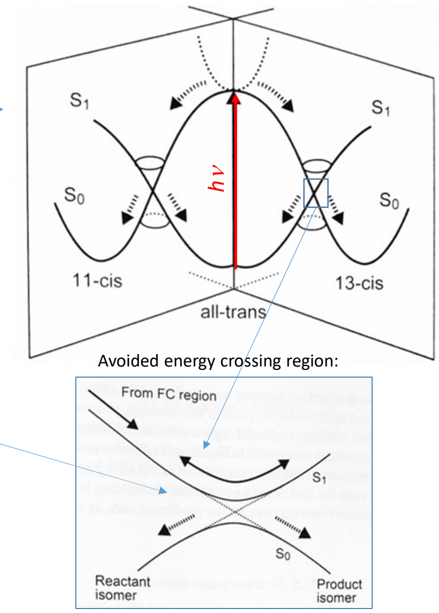
Photoisomerization

Example: bacterial rhodopsin, all-trans to 13-cis transition

Potential energy profiles governing photoisomerization



Initial state: all-trans conformation

Excitation with $h\nu$: vertical (FC) transition to the excited all-trans state.

From the top, it can fall toward 11-cis or 13-cis. Protein ensures that there is a slope towards 13-cis, so it does not end up in 11-cis

Once in the avoided energy crossing region, it can fall back to all-trans conformation, or to 13-cis.

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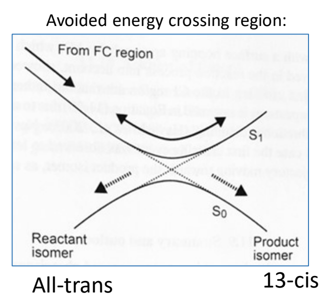
Photoisomerization

Example: bacterial rhodopsin, all-trans to 13-cis transition



In the conical intersection region:
First pass (→) – due to kinetic motion energy it will be crossing to the product (13-cis)
If not crossing in the first oscillation, it will come back (moving ←) and cross back to the reactant (all-trans), etc.

Assuming constant transition probability θ at each crossing event, can calculate the overall branching ratio Φ=product/reactant after absorbing light:

$$\Phi = (1-f)[\theta + \theta(1-\theta)^2 + \theta(1-\theta)^4 + \cdots] = (1-f)/(2-\theta)$$

Where f is population lost before crossing (fluorescence and non-adiabatic transition, ≈0 for (bacterio)retinal)

$\theta \to 1$ then Φ=1 – all will fall into the product at the first pass of AEC
$\theta \to 0$ then $\Phi \to 0.5$ – equal probability to fall in each direction

Measured Φ>0.6 for bR and Rh → θ>0.33, i.e., ~3 passes are sufficient for crossing most of the reactant into products

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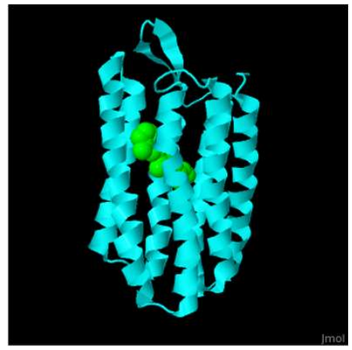
EXTRA

Movie: https://youtu.be/r6v21W8zRIw



https://proteopedia.org/wiki/index.php/Bacteriorhodopsin

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EXTRA

$$\hat{T}_N\left(\vec{R}\right) = \sum_A^{N_{nuc}}\left(-\frac{\hbar^2}{2M_A}\nabla_A^2\right)$$

Integration over *r*, Φ(R) is "constant"

$$\left\langle \Phi_J\left(\vec{R}\right)\psi_J\left(\vec{r},\vec{R}\right)\middle|\hat{T}_N\left(\vec{R}\right)+\hat{H}_{BO}\left(\vec{r},\vec{R}\right)\middle|\Phi_I\left(\vec{R}\right)\psi_I\left(\vec{r},\vec{R}\right)\right\rangle_r = \Phi_J\left\langle\psi_J\middle|\hat{T}_N\middle|\Phi_I\psi_I\right\rangle_r + \Phi_J\left\langle\psi_J\middle|\hat{H}_{BO}\middle|\Phi_I\psi_I\right\rangle_r$$

$$\Phi_J\left\langle\psi_J\middle|\hat{H}_{BO}\middle|\Phi_I\psi_I\right\rangle_r = \Phi_I\Phi_J\left\langle\psi_J\middle|\hat{H}_{BO}\middle|\psi_I\right\rangle_r = \Phi_I\Phi_J E_I\left\langle\psi_J\,|\,\psi_I\right\rangle_r = 0$$

$$\Phi_J\left\langle\psi_J\middle|\hat{T}_N\middle|\Phi_I\psi_I\right\rangle_r = \Phi_J\left\langle\psi_J\middle|\sum_A^{N_{nuc}}\left(-\frac{\hbar^2}{2M_A}\nabla_A^2\right)\middle|\Phi_I\psi_I\right\rangle_r =$$

Drop second derivatives (small curvature)

$$= -\Phi_J\sum_A^{N_{nuc}}\left\langle\psi_J\middle|\frac{\hbar^2}{2M_A}\nabla_A^2\middle|\Phi_I\psi_I\right\rangle_r = -\Phi_J\sum_A^{N_{nuc}}\frac{\hbar^2}{2M_A}\left\langle\psi_J\,|\,\psi_I\nabla_A^2\Phi_I + 2\nabla_A\Phi_I\nabla_A\psi_I + \Phi_I\nabla_A^2\psi_I\right\rangle_r =$$

$$\simeq -\Phi_J\sum_A^{N_{nuc}}\frac{\hbar^2}{2M_A}\left\langle\psi_J\,|\,2\nabla_A\Phi_I\nabla_A\psi_I\right\rangle_r = -\Phi_J\sum_A^{N_{nuc}}\frac{\hbar^2}{2M_A}\left\langle\psi_J\,|\,2\nabla_A\Phi_I\nabla_A\psi_I\right\rangle_r = -i\hbar\Phi_J\sum_A^{N_{nuc}}\left\langle\psi_J\middle|\nabla_A\middle|\psi_I\right\rangle_r\frac{\hbar}{iM_A}\nabla_A\Phi_I$$

Drop Φ's on right and left:

$$\left\langle\psi_J\left(\vec{r},\vec{R}\right)\middle|\hat{T}_N\left(\vec{R}\right)\middle|\psi_I\left(\vec{r},\vec{R}\right)\right\rangle_r = -i\hbar\sum_A^{N_{nuc}}\left\langle\psi_J\middle|\nabla_A\middle|\psi_I\right\rangle_r\frac{\hbar}{iM_A}\nabla_A$$

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EXTRA

Follow the same approach as for energy transfer between two states.

Born-Oppenheimer approximation:

- Assume that electronic wavefunction depends on the positions of the nuclei
- Separate Hamiltonian into nuclear kinetic energy (T) and Born-Oppenheimer Hamiltonian (H_{BO}):

$$\hat{H}(\mathrm{r},\mathrm{R}) = \hat{T}_N(\mathrm{R}) + \hat{H}_{BO}(\mathrm{r},\mathrm{R})$$ ← **R, r** – coordinates of nuclei and electrons

Where:

$$\hat{T}_N(\vec{R}) = \sum_a^{N_{nuc}} \left(-\frac{\hbar^2}{2M_a} \nabla_a^2 \right)$$

Kinetic energy of N_{nuc} nuclei, each of mass M_a

$$\hat{H}_{BO}(\mathbf{r},\mathbf{R}) = \underbrace{\sum_i^{N_{el}} \left(-\frac{\hbar^2}{2m} \nabla_i^2 \right)}_{\text{Kinetic energy of electrons } (m=m_e)} - \underbrace{\sum_i^{N_{el}} \sum_a^{N_{nuc}} \frac{Z_a e^2}{r_{ia}}}_{\text{Energy of electron-nuclei attraction}} + \underbrace{\sum_i^{N_{el}} \sum_{j>i}^{N_{el}} \frac{e^2}{r_{ia}}}_{\text{Energy of electron-electron repulsion}} + \underbrace{\sum_a^{N_{nuc}} \sum_{b>a}^{N_{nuc}} \frac{Z_a Z_b e^2}{r_{ab}}}_{\text{Energy of nuclei repulsion}}$$

Separate variables in the wavefunctions of two states (ground and excited, subscripts i=0,1):

$$\Psi_i(t,\mathrm{r},\mathrm{R}) = \psi_i(\mathrm{r};\mathrm{R})\chi_i(\mathrm{r},\mathrm{R})\mathrm{e}^{-iE_i^{BO}(\mathrm{R})t/\hbar}$$ ← ψ_0 depends only *parametrically* on **R**

Where spatial electronic wavefunctions ψ are solutions to:

$$\hat{H}_{BO}(\mathbf{r},\mathbf{R})\psi_i(\mathbf{r};\mathbf{R}) = E_i^{BO}(\mathbf{R})\psi_i(\mathbf{r};\mathbf{R})$$

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EXTRA

$$\hat{H}(\mathrm{r},\mathrm{R}) = \hat{T}_N(\mathrm{R}) + \hat{H}_{BO}(\mathrm{r},\mathrm{R}) \qquad \Psi_i(t,\mathrm{r},\mathrm{R}) = \psi_i(\mathrm{r};\mathrm{R})\chi_i(\mathrm{r},\mathrm{R})\mathrm{e}^{-iE_i^{BO}(\mathrm{R})t/\hbar}$$

To describe transition between two states consider combination of two states

$$\Psi(t,\mathrm{r},\mathrm{R}) = c_0(t)\Psi_0(t,\mathrm{r},\mathrm{R}) + c_1(t)\Psi_1(t,\mathrm{r},\mathrm{R})$$

And solve time-dependent Schrödinger equation:

$$i\hbar\frac{\partial\Psi}{\partial t} = \mathrm{H}\Psi$$

$$\frac{\partial}{\partial t}\left\{c_0(t)\Psi_0(t,\mathrm{r},\mathrm{R}) + c_1(t)\Psi_1(t,\mathrm{r},\mathrm{R})\right\} = -i\hbar^{-1}\hat{H}\left\{c_0(t)\Psi_0(t,\mathrm{r},\mathrm{R}) + c_1(t)\Psi_1(t,\mathrm{r},\mathrm{R})\right\}$$

As usual, multiply with Ψ_1 and integrate over all coordinates **R,r**:

Left side:

$$\left\langle \Psi_1 \middle| \frac{\partial}{\partial t}\{c_0\Psi_0 + c_1\Psi_1\} \right\rangle = \left\langle \Psi_1 \middle| \frac{\partial}{\partial t} c_0\Psi_0 \right\rangle + \left\langle \Psi_1 \middle| \frac{\partial}{\partial t} c_1\Psi_1 \right\rangle$$

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EXTRA

$$\hat{H}(\mathrm{r},\mathrm{R}) = \hat{T}_N(\mathrm{R}) + \hat{H}_{BO}(\mathrm{r},\mathrm{R}) \qquad \Psi_i(t,\mathrm{r},\mathrm{R}) = \psi_i(\mathrm{r};\mathrm{R})\,\chi_i(\mathrm{r},\mathrm{R})\,\mathrm{e}^{-iE_i^{BO}(\mathrm{R})t/\hbar}$$

$$\left\langle \Psi_1 \middle| \frac{\partial}{\partial t}\{c_0\Psi_0 + c_1\Psi_1\} \right\rangle = \left\langle \Psi_1 \middle| \frac{\partial}{\partial t} c_0\Psi_0 \right\rangle + \left\langle \Psi_1 \middle| \frac{\partial}{\partial t} c_1\Psi_1 \right\rangle$$

$$= \left\langle \Psi_1 \middle| \frac{\partial c_0}{\partial t}\Psi_0 \right\rangle + \left\langle \Psi_1 \middle| c_0 \frac{\partial \Psi_0}{\partial t} \right\rangle + \left\langle \Psi_1 \middle| \frac{\partial c_1}{\partial t}\Psi_1 \right\rangle + \left\langle \Psi_1 \middle| c_1 \frac{\partial \Psi_1}{\partial t} \right\rangle =$$

$$= \frac{\partial c_0}{\partial t}\langle \Psi_1 | \Psi_0 \rangle + c_0 \left\langle \Psi_1 \middle| \frac{\partial \Psi_0}{\partial t} \right\rangle + \frac{\partial c_1}{\partial t}\langle \Psi_1 | \Psi_1 \rangle + c_1 \left\langle \Psi_1 \middle| \frac{\partial \Psi_1}{\partial t} \right\rangle =$$

$$= c_0 \left\langle \Psi_1 \middle| \frac{\partial \Psi_0}{\partial t} \right\rangle + \frac{\partial c_1}{\partial t} + c_1 \left\langle \Psi_1 \middle| \frac{\partial \Psi_1}{\partial t} \right\rangle$$

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EXTRA

$$\hat{H}(\mathrm{r},\mathrm{R}) = \hat{T}_N(\mathrm{R}) + \hat{H}_{BO}(\mathrm{r},\mathrm{R}) \qquad \Psi_i(t,\mathrm{r},\mathrm{R}) = \psi_i(\mathrm{r};\mathrm{R})\,\chi_i(\mathrm{r},\mathrm{R})\,\mathrm{e}^{-iE_i^{BO}(\mathrm{R})t/\hbar}$$

$$\frac{\partial}{\partial t}\{c_0(t)\Psi_0(t,\mathrm{r},\mathrm{R}) + c_1(t)\Psi_1(t,\mathrm{r},\mathrm{R})\} = -i\hbar^{-1}\hat{H}\{c_0(t)\Psi_0(t,\mathrm{r},\mathrm{R}) + c_1(t)\Psi_1(t,\mathrm{r},\mathrm{R})\}$$

Right side: $$\left\langle \Psi_1 \middle| \hat{H} \middle| c_0\Psi_0 + c_1\Psi_1 \right\rangle = c_0\left\langle \Psi_1 \middle| \hat{H} \middle| \Psi_0 \right\rangle + c_1\left\langle \Psi_1 \middle| \hat{H} \middle| \Psi_1 \right\rangle$$

$$= c_0\left\langle \psi_1\chi_1\,\mathrm{e}^{-iE_1^{BO}(\mathrm{R})t/\hbar} \middle| \hat{H} \middle| \psi_0\chi_0\,\mathrm{e}^{-iE_0^{BO}(\mathrm{R})t/\hbar} \right\rangle + c_1\left\langle \psi_1\chi_1\,\mathrm{e}^{-iE_1^{BO}(\mathrm{R})t/\hbar} \middle| \hat{H} \middle| \psi_1\chi_1\,\mathrm{e}^{-iE_1^{BO}(\mathrm{R})t/\hbar} \right\rangle$$

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EXTRA

$$\left\langle \Psi_J \left| \left[\nabla, H \right] \right| \Psi_I \right\rangle = ?$$

$$\langle \Psi_J(\mathbf{r}, \mathbf{R}) | \nabla_A | \Psi_I(\mathbf{r}, \mathbf{R}) \rangle_r = \frac{\langle \Psi_J(\mathbf{r}, \mathbf{R}) | [\nabla_A, \ \hat{H}_{BO}(\mathbf{r}, \mathbf{R})] | \Psi_I(\mathbf{r}, \mathbf{R}) \rangle_r}{E_I^{BO}(\mathbf{R}) - E_J^{BO}(\mathbf{R})}$$

$$\left\langle \Psi_J \left| \nabla H \right| \Psi_I \right\rangle = \left\langle \Psi_J \left| \nabla \right| E_I \Psi_I \right\rangle = \left\langle \Psi_J \,|\, \nabla E_I \Psi_I + E_I \nabla \Psi_I \right\rangle = \left\langle \Psi_J \,|\, \Psi_I \right\rangle \nabla E_I + \left\langle \Psi_J \left| \nabla \right| \Psi_I \right\rangle E_I$$

$$\left\langle \Psi_J \left| H \nabla \right| \Psi_I \right\rangle = \left\langle \Psi_J \left| H \right| \nabla \Psi_I \right\rangle = \left\langle \Psi_J \left| \nabla \right| \Psi_I \right\rangle E_I$$

$$H \nabla \left(\Psi_J^* \Psi_I \right) = H (\nabla \Psi_J^* \Psi_I + \Psi_J^* \nabla \Psi_I) = -\frac{\hbar^2}{2m} \nabla_e \nabla_e (\nabla \Psi_J^* \Psi_I + \Psi_J^* \nabla \Psi_I)$$

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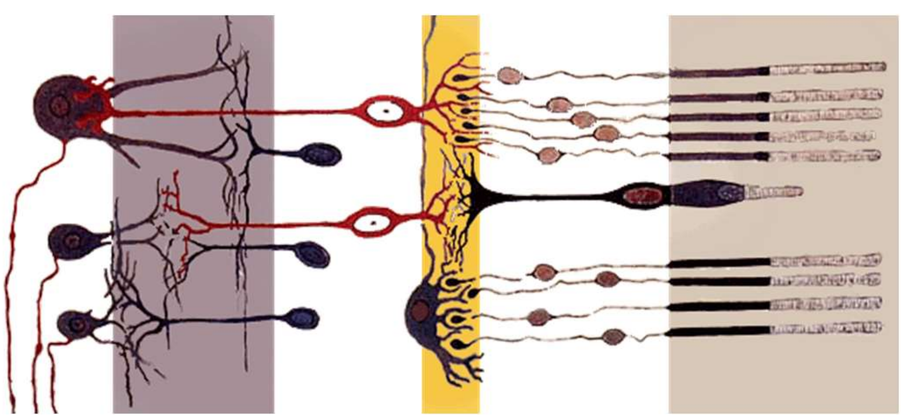
An eye



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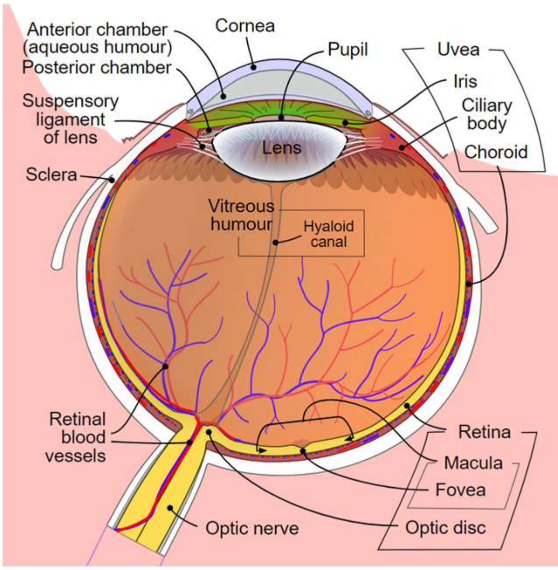
An eye



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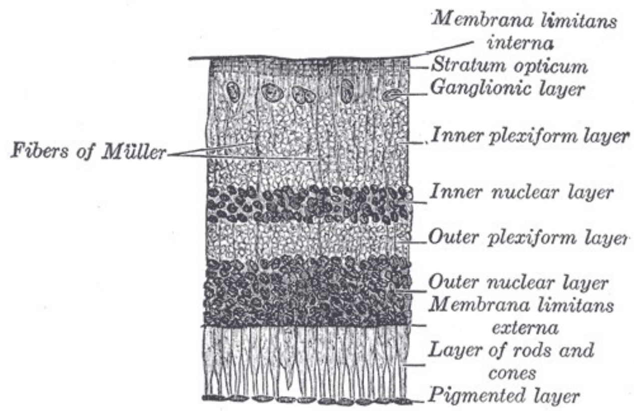

By Henry Vandyke Carter - Henry Gray (1918) Anatomy of the Human Body, p. 1,016 (See "Book" section below)Bartleby.com: Gray's Anatomy, Plate 881, Public Domain, https://commons.wikimedia.org/w/index.php?curid=566820

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